

Global Business Intelligence & Customer Analytics Platform

*Updated project outline with detailed web app dashboard purpose, users, and presentation plan
Built on the Olist E-commerce Dataset*

What this document adds

This revision expands the project outline with a dedicated interactive web app dashboard section. It explains who will use the dashboard, what business decisions it supports, which pages it should include, how the app should be presented, and how it complements Power BI.

1. Project summary

This project is an end-to-end business intelligence platform built on the Olist e-commerce dataset. It combines Python, SQL, Excel, Power BI, and a Streamlit-based interactive web dashboard to analyze sales, customers, sellers, payments, logistics, and customer satisfaction.

The goal is not only to produce charts, but to turn distributed operational data into a decision-making system that different business teams can actually use.

2. Why the web app dashboard matters

The web app dashboard is the interactive front end of the project. While Python and SQL prepare the data and Power BI provides formal BI reporting, the web app gives users a simple interface to explore the business in real time.

- It lets users filter data dynamically by date, state, product category, seller, payment type, review score, and delivery status.
- It gives fast answers without the user needing to understand SQL queries or raw CSV files.
- It supports drill-down analysis, so a manager can move from a company-wide KPI to a specific state, category, seller, or order pattern.
- It makes the project portfolio-ready because it behaves like a real analytics product instead of a static assignment.

Simple explanation

Think of the web app as the decision screen of the project. The backend does the analysis; the dashboard is where people interact with the results.

3. What the web app will be used for

User group	What they check	Business value
Management / leadership	Revenue, orders, growth, delayed delivery rate, average review score	Quick view of business health and where intervention is needed
Sales / category team	Top categories, top sellers, high-revenue states, average order value	Find growth opportunities and focus on winning products or regions
Operations / logistics	Delivery time, delayed orders, freight trends, state-wise performance	Identify bottlenecks and improve shipping efficiency
Customer experience team	Review score distribution, low-rating patterns, delivery delay impact	Understand why customers are unhappy and what to improve
Analyst / project reviewer	Filters, drill-downs, downloadable data, summary KPIs	Validate findings and show that the project is interactive and decision-oriented

4. How the web app should be presented

The web app should be presented as an interactive business portal, not just a collection of charts. Start with a clean landing page that explains the project purpose and the main KPIs being tracked.

Use a left sidebar for filters so the main content area stays clean. The first thing users should see is a row of KPI cards, followed by the most important trend and comparison charts.

Each page should answer a business question. This makes the dashboard feel intentional and easier to explain in presentations, interviews, and portfolio reviews.

- Recommended presentation flow:
 - Open with an Executive Overview page.
 - Show how filters change KPIs in real time.
 - Move into one focused page at a time: customers, product & seller, operations, satisfaction.
 - End with a summary panel or insight box that explains the main findings and possible actions.

5. Recommended web app structure

5.1 Executive Overview

This is the main summary page for leadership and first-time viewers. It should answer: How is the business performing overall?

- KPI cards: Total Revenue, Total Orders, Unique Customers, Average Order Value, Average Review Score, Delayed Order Percentage.
- Charts: revenue trend by month, orders trend by month, revenue by state, revenue by category.
- Use: gives a one-screen health check of the business.

5.2 Customer Analytics

This page explains customer behavior and customer value. It should answer: Who are the most important customers and where do they come from?

- KPIs: total unique customers, repeat customer rate, average spend per customer.
- Charts: customers by state, repeat vs one-time customers, top customers by spend, order frequency distribution.
- Use: supports retention strategy and customer targeting.

5.3 Product & Seller Performance

This page focuses on what is sold and who is selling it. It should answer: Which categories and sellers are driving the business?

- KPIs: top category by revenue, average revenue per seller, best seller revenue.
- Charts: top categories, top sellers, seller performance by state, freight by category.
- Use: helps identify strong and weak commercial areas.

5.4 Operations & Delivery

This page focuses on shipping performance and process efficiency. It should answer: Where are delivery problems happening?

- KPIs: average delivery days, on-time rate, delayed order percentage, average freight value.
- Charts: delivery time distribution, delayed orders by state, delivery days by category, seller performance vs delivery time.
- Use: helps improve logistics and operational planning.

5.5 Customer Satisfaction

This page explains review outcomes and customer experience. It should answer: What drives good or bad reviews?

- KPIs: average review score, low-review percentage.
- Charts: review score distribution, review score vs delay, review score by category, review score by payment type.
- Use: connects operational performance to customer satisfaction.

6. Filters and interactions to include

- Date range selector
- Customer state
- Seller state
- Product category
- Payment type
- Review score band
- Delayed vs non-delayed orders
- Optional seller selector for detailed drill-down

These filters should update the KPI cards, charts, and tables instantly so the app feels genuinely interactive.

7. Features that make the dashboard valuable

- **KPI cards:** Give fast, high-level answers before the user reads any chart.
- **Interactive charts:** Let users hover, zoom, and inspect details without leaving the page.
- **Downloadable filtered data:** Shows practical value because users can export the current slice for reporting or further analysis.
- **Insight callouts:** A short text block under major charts should explain what the user is seeing and why it matters.
- **Navigation by page:** Keeps the app clear and prevents one page from becoming overcrowded.

8. How it fits with the rest of the project

- Python prepares and cleans the raw Olist files, creates derived fields such as delivery days, delayed flag, order month, and translated product category names.
- SQL stores and organizes the cleaned data into analysis-ready tables or views.
- Power BI delivers polished business dashboards for formal reporting.
- The web app delivers flexible exploration and portfolio-ready interactivity.

9. Suggested tech stack for the web app

- Streamlit for the app interface
- Pandas for in-app filtering and calculations
- Plotly for interactive charts
- PostgreSQL or cleaned CSV files as the data source
- Optional CSS styling inside Streamlit for cleaner UI
- GitHub + Streamlit Community Cloud for publishing

10. Suggested presentation script

1. Start by explaining that the dashboard is designed for managers and operational teams to explore business performance interactively.
2. Show the Executive Overview page first and point out the headline KPIs.
3. Use one filter, such as product category or state, to demonstrate live interaction.
4. Move to Operations & Delivery and explain how delays can be tied to customer reviews.
5. Finish by explaining that the web app complements Power BI by giving a more exploratory, product-like interface.

11. Final deliverables

- Updated Word document and PDF outline
- Python cleaning notebook
- SQL schema and query file
- Power BI dashboard
- Interactive Streamlit web app
- README / GitHub documentation